Station	Parameter	Description	Comments
Freestream			
	Alt (kft)	Altitude	
	M	Mach number	
	P (psf)	Static pressure	
	Pt (psf)	Total pressure	
	T (R)	Static temperature	
	Tt (R)	Total temperature	
	q (psf)	Flight dynamic pressure	Used for Dn calculation
	Cdo	Cowl/nacelle drag coefficient	
	Ao (ft2)	Freestream area of propulsion streamtube	
	U (ft/s)	Freestream velocity	
	md (sl/s)	Propulsion airflow	
	An (ft2)	Area of cowl/nacelle	
	Dn (lbf)	Cowl/nacelle drag	
	Dram (lbf)	Ram drag	
Constants			
	Tref	Reference temperature	Not used in this implementation.
	gamma	Ratio of heat capacities	
	Δhf	Heat of combustion	
	a	Coefficient for combustion efficiency fit	
	b	Coefficient for combustion efficiency fit	
Inlet			
	Π_d	Inlet recovery (Pt2/Pt0)	
	Pt (psf)	Inlet total pressure	
	Tt (R)	Inlet total temperature	
	md (sl/s)	Inlet airflow	
	phi	Air equivalence ratio	Always zero. Use in "MassFracs" function.
	beta	Air nitrogen-to-oxygen molar ratio	Always 3.762. Use in "MassFracs" function.
	md H2 (sl/s)	Mass flow of hydrogen in captured air stream	Zero.
	md O2 (sl/s)	Mass flow of oxygen in captured air stream	
	md N2 (sl/s)	Mass flow of nitrogent in captured air stream	
	Mb (g/mol)	Mixture molecular weight	From MWT_yHyOyN function.
	R (ft2/s2.R)	Mixture gas constant	
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	
Fan			
	Π_f	Fan pressure ratio	
	η_f	Fan isentropic efficiency	
	Pt (psf)	Total pressure after the fan	
	Tt (R)	Total temperature after the fan	
	md (sl/s)	Mass flow of air through the fan	
	Pwr (ft-lbf/s)	Power consumed by the fan	
	md H2 (sl/s)	Mass flow of hydrogen in fan flow	Still zero.
	md O2 (sl/s)	Mass flow of oxygen in fan flow	Same as in inlet flow.
	md N2 (sl/s)	Mass flow of nitrogent in fan flow	Same as in inlet flow.
	Mb (g/mol)	Mixture molecular weight	Same as in inlet flow.
	R (ft2/s2.R)	Mixture gas constant	Same as in inlet flow.
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	Same as in inlet flow.
Station 15			
	M	Mach number	
	Pt (psf)	Total pressure	
	P (psf)	Static pressure	
	Tt (R)	Total temperature	
	T (R)	Static temperature	
	md (sl/s)	Mass flow of air entering the ramburner	
	A (ft2)	Flow area of air entering the ramburner	
	U (ft/s)	Flow velocity	
	I (lbf)	Air impulse entering the ramburner	
	Ht (ft.lb/s)	Flow rate of total enthalpy of air	For reference.
	md H2 (sl/s)	Mass flow of hydrogen in the airflow entering ramburner	Still zero.

Station	Parameter	Description	Comments
	md O2 (sl/s)	Mass flow of oxygen in the airflow entering ramburner	Same as in inlet flow.
	md N2 (sl/s)	Mass flow of nitrogen in the airflow entering ramburner	Same as in inlet flow.
	Mb (g/mol)	Mixture molecular weight	Same as in inlet flow.
	R (ft2/s2.R)	Mixture gas constant	Same as in inlet flow.
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	Same as in inlet flow.
Dummy Calc for Ttb			
	Tt1 (R)	Propellant feed tempertature	
	Pt (psf)	Chamber pressure	
	md ch (sl/s)	Chamber mass flow rate	Arbitrary to compute chamber Ttb.
	phi	Chamber equivalence ratio	Vary as requested. Use in "MassFracs" function.
	beta	Chamber nitrogen-to-oxygen molar ratio	Always zero. Use in "MassFracs" function.
	Cp_H2 (ft2/s2.R)	Heat capacity at constant pressure for hydrogen	
	Cp_O2 (ft2/s2.R)	Heat capacity at constant pressure for oxygen	
	md H2 (sl/s)	Mass flow of hydrogen	Based on arbitrary mass flow value.
	md O2 (sl/s)	Mass flow of oxygen	Based on arbitrary mass flow value.
	md N2 (sl/s)	Mass flow of nitrogen	
	m_brn H2 (sl/s)	Amount of hydrogen consumed	For reference. Does not need to be computed.
	Mbp (g/mol)	Mixture molecular weight	
	R (ft2/s2.R)	Mixture gas constant	
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	
	ht_i (ft2/s2)	Initial enthalpy of flows entering the chamber control volume	
	η_b	Combustion efficiency	For reference.
	Ttb (R)	Mixture temperature after combustion	
Turbine Exit			
	η_t	Turbine efficiency	
	η_m	Shaft mechanical efficiency	
	Mx	Primary exit Mach number	
	Px (psf)	Primary exit static pressure	
	Ptx (psf)	Turbine/primary exit total pressure	
	Tx (R)	Primary exit static temperature	
	Ttx (R)	Turbine/primary exit total temperature	
	dh (ft2/s2)	Reduction in specific enthalpy across the turbine	
	md (sl/s)	Turbine/primary mass flow entering the ramburner	Needed to meet fan power requirements.
	phi	Chamber equivalence ratio	To recompute mass fractions (could use chamber values).
	beta	Chamber nitrogen-to-oxygen molar ratio	To recompute mass fractions (could use chamber values).
	md H2 (sl/s)	Mass flow of hydrogen	Actual value, scaled to meet power requirements.
	md O2 (sl/s)	Mass flow of oxygen	Actual value, scaled to meet power requirements.
	md N2 (sl/s)	Mass flow of nitrogen	Still zero.
	m_brn H2 (sl/s)	Amount of hydrogen consumed	For reference. Does not need to be computed.
	Mbp (g/mol)	Mixture molecular weight	Should be the same as the dummy calculation.
	R (ft2/s2.R)	Mixture gas constant	Should be the same as the dummy calculation.
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	Should be the same as the dummy calculation.
	Ux (ft/s)	Primary exit velocity	
	Ax (ft2)	Primary exit area entering the ramburner	
	I (lbf)	Primary exit impulse entering the ramburner	
Station 7			
	md (sl/s)	Combined mass flow	Used in Mfi.
	md H2	Mass flow of all hydrogen atoms	
	md O2	Combined mass flow of all oxygen atoms	
	md N2	Mass flow of all nitrogen atoms	
	Mb (g/mol)	Mixture molecular weight	
	R (ft2/s2.R)	Mixture gas constant	
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	
	ht_i (ft2/s2)	Initial enthalpy of flows entering the propulsion control volume	
	η_b	Combustion efficiency	For reference.
	Tt (R)	Mixture temperature after combustion	Tt burned
	Ht (ft.lb/s)	Flow rate of total enthalpy of combusted mixture	Based on Tt burned. Used in Mfi.
	I (lbf)	Combined impulse of the airflow and turbine exhaust	Used in Mfi.
	A (ft2)	Combined area of the airflow and turbine exhaust	
	rho (sl/ft3)	Flow density	

Station	Parameter	Description	Comments
	U (ft/s)	Flow velocity	
	T (R)	Static temperature	
	M	Mach number	From Mfi.
	P (psf)	Static pressure	
	Pt (psf)	Total pressure	
Nozzle			
	Mb	Mixture molecular weight	From station 7.
	R (ft2/s2.R)	Mixture gas constant	
	Cp (ft2/s2.R)	Mixture heat capacity at constant pressure	
	Tt (R)	Total temperature	From station 7.
	T (R)	Static temperature	
	Pt (psf)	Total pressure	From station 7.
	P (psf)	Static pressure	Expanded to ambient pressure.
	M	Mach number	
	A (ft2)	Flow area	
	U (ft/s)	Flow velocity	
	md (sl/s)	Mass flow rate	
	Cfg	Nozzle gross thrust coefficient	
	F (lbf)	Nozzle gross thrust	
Performance			
	FN (lbf)	Net propulsive force	Fg - Dram - Dn
	SFCeng (lbm/hr.lbf)	Specific fuel consumption	
	FNWA (lbf.s/sl)	Airflow specific thrust	Thrust per slug of airflow. (Usually per pound of airflow.)
	FN/Amx	Thrust loading (per propulsion projected area)	Thrust per Ao.
	ISP (s)	Specific impulse	